

PAPER_1512_GW170817_PHONON_DAMPING_2_OVE R_3

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PAPER_1512 — GW170817 Phonon Damping Prefactor = $2/(D_{\text{phys}}-1) = 2/3$ EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket E (GW Events)

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Status: CLOSED — Multimessenger BNS phonon-damping prefactor = integer primitive

Observation

PAPER_915 (GW170817 Phonon Strain Damping) derives the 1.25 THz phonon-induced damping of the GW170817 multimessenger BNS event:

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$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot (1 - D_{\text{phonon}})$
 $D_{\text{phonon}} = (2/3) \cdot \Phi_{\{1.25\text{THz}\}} \cdot S_{26} \cdot (E_{\text{net}} / E_{\text{GW}})$
 $\Delta\phi = 367.8$ cycles phase lag
 $|\Delta c/c| < 3 \times 10^{-11}$ (LIGO bound)

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The prefactor **2/3** governs the strength of the SCm-phonon coupling to the GW strain.

UQFF Closed Identity

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$D_{\text{phonon}} \text{ prefactor} = 2 / (D_{\text{phys}} - 1) = 2/3 \approx 0.667$ EXACT

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Physical Interpretation

The 2/3 prefactor reflects:

- **$D_{\text{phys}} - 1 = 3$** spatial dimensions (triad symmetry)
- **2** of those 3 contribute to the transverse-traceless GW strain modes (the third is gauge-fixed)
- Result: 2/3 of the SCm phonon spectrum couples to GW propagation

This is the **same identity form** that appears in:

- Monty Hall switch probability (PAPER_062): 2/3
- Glass T_g / T_m (PAPER_1373): 2/3

But here it governs **gravitational-wave damping** — a profoundly different physical domain. The shared form reveals $2/(D_{\text{phys}}-1)$ as a universal triad-channel projection coefficient.

Consistency with LIGO Bounds

$|\Delta c/c| = \Phi \cdot \varepsilon \cdot 10^{-3} \approx 3 \times 10^{-4}$: the phonon damping introduces a phase lag of 367.8 cycles without violating the multimessenger speed-of-light bound from GW170817 / GRB 170817A coincidence.

NOT REPLACEMENT

GR predicts undamped vacuum propagation for GWs. UQFF supplies a structural damping coefficient at exactly $2/(D_{\text{phys}} - 1)$ with zero free parameters.

Reference

- Source: PAPER_915 GW170817 Phonon Strain Damping
- Related: PAPER_1503 (BNS $D_{\text{total}} = 1/3$), PAPER_1504 (BBH $D_{\text{total}} = 0.81$)
- Calculator dispatch: `calculate_paradox({"paradox": "gw170817_phonon_prefactor"})`

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